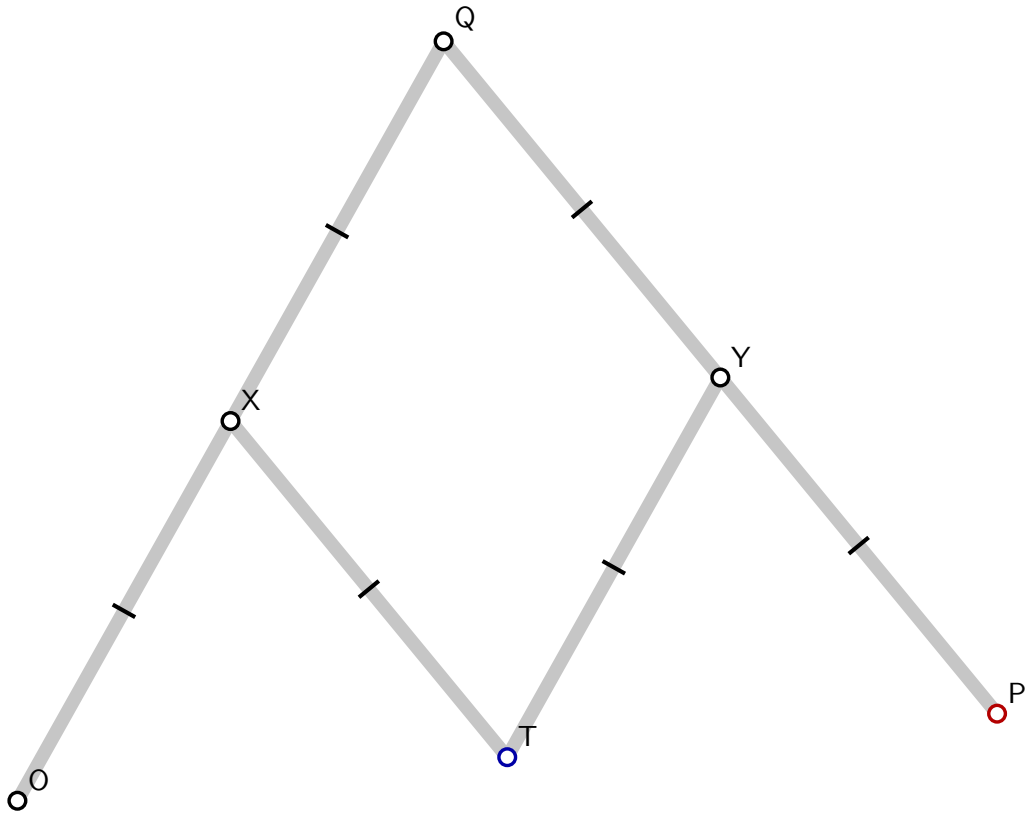
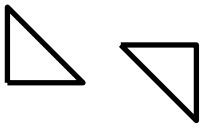
Use the assembled machine and joined mat. Keep the four-sided frame open and uncrossed. Each new copy goes on a fresh clear overlay. An adult resets the anchor or swaps the tips when a problem calls for it.

Problem 1: The six marked lengths are equal. Must P always be twice as far from O as T? Explain using the joined bars.



One square piece is one square unit. Two matching triangle pieces make one square unit.

Two halves



Join edges

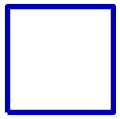


1 square unit

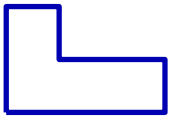


Problem 2: Use inserts G, H and I with the square and triangle pieces. How many square units cover each original and its copy? Find a rule for all three.

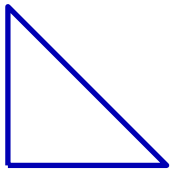
G



H

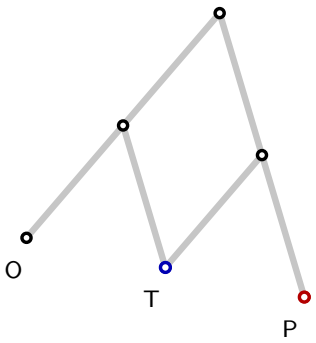


I

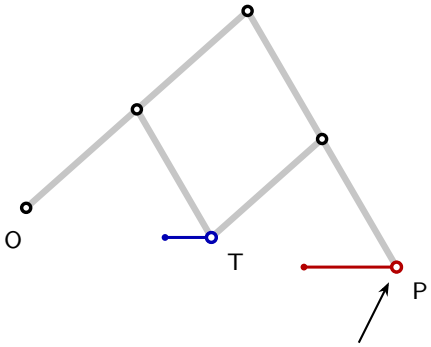


For this page, move P while T draws. Keep O fixed.

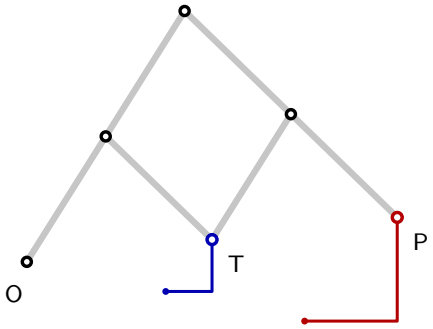
Start



Move P



Copy

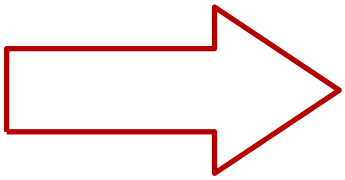


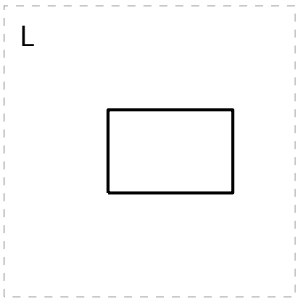
Problem 3: Trace J and K with P. What copies does T make? Find a rectangle for P that makes T follow the small solid box on insert L.

J



K





Problem 4: Trace triangle M with anchor O, then with anchor O′, keeping M in the same place. What changes in the two copies? Find a new place for M that makes the second copy land exactly on the first.

Move T; P draws. The adult moves the anchor between copies.

